

ABBOTTABAD KAKUL, Pakistan

WMO: 415350

Lat: 34.183N

Lon: 73.250E

Elev: 4295

StdP: 12.55

Time Zone: 5.00 (E05)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 30.9	(c) 32.8	(d) 20.5	(e) 18.0	(f) 44.9	(g) 22.8	(h) 20.1	(i) 44.6	(j) 4.4	(k) 53.6	(l) 3.3	(m) 53.0	(n) 0.0	(o) N/A	(p) 0.315

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 23.4	(c) 93.4	(d) 68.5	(e) 90.9	(f) 69.0	(g) 88.4	(h) 69.4	(i) 76.3	(j) 84.8	(k) 75.3	(l) 83.6	(m) 74.3	(n) 82.3	(o) 1.7	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	149.1	81.7	73.0	144.0	80.7	71.9	138.8	79.5	43.5	85.7	42.3	83.2	41.3	83.5	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.1	(b) 3.2	(c) 3.0	(e) 28.5	(f) 97.9	(g) 1.5	(h) 2.1	(i) 27.4	(j) 99.5	(k) 26.5	(l) 100.7	(m) 25.7	(n) 101.9	(o) 24.6	(p) 103.5		
(a) 4.1	(b) 3.2	(c) 3.0	(e) 25.5	(f) 78.6	(g) 1.7	(h) 1.5	(i) 24.3	(j) 79.7	(k) 23.3	(l) 80.6	(m) 22.4	(n) 81.5	(o) 21.1	(p) 82.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	44.7	47.3	55.0	63.5	71.4	76.7	76.8	75.4	71.9	64.5	55.0	48.0	(6)
(7)		DBStd	12.43	4.23	4.87	6.46	5.91	4.76	4.14	3.01	2.85	3.12	4.84	4.13	3.90	(7)
(8)		HDD50	393	170	101	26	1	0	0	0	0	0	0	8	87	(8)
(9)		HDD65	2441	628	496	316	95	8	0	0	0	2	68	300	528	(9)
(10)		CDD50	4989	7	25	179	406	664	801	831	787	657	450	158	24	(10)
(11)		CDD65	1564	0	0	5	50	207	351	366	323	209	53	0	0	(11)
(12)		CDH74	14278	0	0	77	615	2235	3865	3132	2350	1489	504	11	0	(12)
(13)		CDH80	5014	0	0	4	145	886	1839	1118	630	323	69	0	0	(13)
(14)	Wind	WSAvg	0.6	0.5	0.6	0.7	0.7	0.8	0.8	0.7	0.7	0.6	0.5	0.4	0.3	(14)
(15)	Precipitation	PrecAvg	52.4	2.8	4.5	4.9	4.4	3.0	3.8	9.8	10.1	3.9	2.0	1.2	1.8	(15)
(16)		PrecMax	63.6	6.7	8.5	7.9	10.0	6.5	14.8	19.5	14.7	8.1	6.8	3.7	5.1	(16)
(17)		PrecMin	40.6	0.0	0.3	0.6	0.4	1.1	0.4	3.3	5.1	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	7.0	1.9	2.2	2.0	2.6	1.5	3.7	3.5	2.6	2.0	1.9	1.3	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.8	70.0	80.0	88.5	94.4	98.4	94.7	89.0	87.1	84.7	75.4	68.1	(19)
(20)			MCWB	46.9	50.5	59.4	64.3	66.2	69.1	71.0	75.3	70.5	66.3	55.0	49.5	(20)
(21)		2%	DB	61.5	64.9	76.1	84.1	91.2	94.8	91.2	87.0	85.0	82.0	71.8	64.6	(21)
(22)			MCWB	45.6	48.3	57.8	62.4	65.7	69.0	72.4	74.3	70.0	64.1	54.0	48.1	(22)
(23)		5%	DB	58.8	61.5	72.6	80.5	88.2	92.2	88.5	85.5	83.5	79.4	69.1	61.6	(23)
(24)			MCWB	44.4	47.3	55.5	61.3	65.2	68.7	72.5	73.9	69.4	62.9	53.2	46.6	(24)
(25)		10%	DB	55.6	58.2	68.8	77.2	85.5	89.5	86.0	83.8	82.0	76.7	66.2	58.7	(25)
(26)			MCWB	43.1	46.2	53.9	59.9	64.3	68.5	73.2	73.4	69.1	60.8	51.5	45.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	53.0	61.7	66.2	69.6	76.3	77.3	78.0	74.9	70.5	58.3	51.2	(27)
(28)			MCDWB	58.0	63.7	74.3	85.2	86.9	87.8	86.4	86.3	83.2	80.6	69.3	63.3	(28)
(29)		2%	WB	47.2	50.8	58.9	63.8	67.6	74.6	76.1	76.4	72.9	68.1	56.3	49.3	(29)
(30)			MCDWB	57.6	61.0	73.3	80.8	86.6	85.8	84.9	84.2	81.8	78.6	68.5	61.0	(30)
(31)		5%	WB	45.6	48.8	56.7	62.3	66.3	73.1	75.1	75.4	71.8	65.0	54.4	47.7	(31)
(32)			MCDWB	56.0	58.6	70.3	77.8	85.5	84.6	83.6	83.1	80.9	75.3	66.4	59.3	(32)
(33)		10%	WB	44.1	47.1	54.7	60.9	64.8	71.6	74.2	74.4	70.5	62.3	52.4	46.2	(33)
(34)			MCDWB	53.6	55.8	66.7	75.2	83.0	83.2	82.3	81.5	79.0	73.7	63.4	57.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.5	19.3	21.2	23.1	24.5	23.4	16.9	16.0	19.7	24.4	23.2	22.4	(35)
(36)			MCDBR	26.1	26.0	27.4	28.3	28.8	27.8	21.9	18.8	22.3	26.3	27.1	26.9	(36)
(37)			MCWBR	13.4	12.7	12.6	12.4	11.0	10.8	9.5	9.0	10.4	11.2	12.6	13.2	(37)
(38)		5% WB	MCDWB	22.4	21.9	25.5	26.2	26.7	20.5	16.8	15.9	18.4	22.9	23.8	24.2	(38)
(39)			MCWBR	12.6	11.9	13.2	12.4	11.2	9.9	8.7	8.5	9.9	11.5	12.4	12.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.371	0.420	0.431	0.444	0.460	0.524	0.588	0.563	0.475	0.426	0.398	0.366	(40)	
(41)		taud	2.129	2.029	2.054	2.072	2.050	1.912	1.816	1.875	2.047	2.078	2.084	2.161	(41)	
(42)		Ebn at Noon	257	256	266	268	265	247	230	233	249	251	244	250	(42)	
(43)		Edh at Noon	38	48	50	52	54	62	67	62	50	45	40	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	886	1039	1436	1807	2175	2231	1964	1784	1724	1484	1079	868	(44)	
(45)		RadStd	112	169	159	126	139	113	148	140	108	91	81	82	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page